OSI HACKATHON 2026 · DEMONSTRATION VIDEO PRODUCTION PACKAGE

Project Negative Lines: 3.5-Minute Demo Video Script & Storyboard

Hand-Annotated Gold Data Reconciliation · Manual Engineering vs. Automated Antigravity AI · Green PFAS Compliance

Target Duration: 3 min 30 sec (210s) | Spoken Words: ~460 words (~132 wpm) | Full 1080p Screencast (16:9) | Dual-Track Streamlit Architecture

PROJECT LINEAGE & TECHNICAL DISTINCTION (CRUCIAL CONTEXT FOR PRESENTERS)

1. Ground Truth Origin: Two researchers (**Akhila** and **Kathe**) independently hand-annotated 10 semiconductor negative-resist patents. Automated discrepancy comparison auditing then cross-verified and resolved conflicting entries (text vs. table conflicts, stoichiometry, units) into a 100% consensus Master Reconciled Gold Set (239 formulation records).

2. Automated Compliance Screening: The reconciled database was evaluated through an automated regulatory screener (OECD PFAS, EU REACH Annex XVII, US TSCA) to isolate persistent fluorinated chemistries.

3. The Two Modeling & Implementation Pathways:

- **Manual Track (Manual Evaluation & Coding):** The Bayesian optimization, surrogate modeling, stoichiometric evaluations, and physics calculations were designed, hand-coded, and manually executed step-by-step by human engineers—giving process experts granular control without reliance on automated black-box loops.
- **Automated Track (Autonomous Antigravity Agent):** The entire end-to-end pipeline—Gaussian Process fitting with a Matérn 5/2 kernel, constrained Expected Improvement acquisition, epistemic uncertainty quantification (±1.96σ), and automated green substitution—was constructed and automated autonomously by **Antigravity**.

DEMO VIDEO STORYBOARD (PART I: INTRODUCTION & DATA FOUNDATION)

SCENE 1: The Nano-Lithography Dilemma & Regulatory PFAS Cliff

`0:00 – 0:35 (35 sec)`

VISUAL & ACTION

Open on full-screen capture of Streamlit portal with **Dark Enterprise Sidebar**.
Mouse hovers over **KPI Metric Cards**: 14 Standardized Records, Target LWR ≤ 4.5 nm, 100% PFAS-Free.
Show hero badge: *OSI Hackathon 2026 · Project Negative Lines*.

SPOKEN VOICEOVER (VO)

"In advanced semiconductor manufacturing, patterning sub-20-nanometer features requires negative-tone photoresists with exceptional line-width roughness and extreme sensitivity.

Historically, achieving this demanded months of trial-and-error e-beam fab trials and heavy reliance on perfluorinated 'forever chemicals'—PFAS. With impending global bans from the OECD and EU REACH, chipmakers face an urgent crisis: eliminate PFAS without sacrificing sub-5-nanometer resolution.

This is **Project Negative Lines**—an end-to-end platform bridging verified empirical data, manual modeling, and autonomous AI optimization."

ON-SCREEN DISPLAY / NOTES

Lower-Third:
Project Negative Lines
AI & Manual Lithography Platform

`LWR ≤ 4.50 nm`

`100% PFAS-Free`

Pacing: Confident, measured tone.

SCENE 2: Hand Annotation to Automated Machine Reconciliation

`0:35 – 1:10 (35 sec)`

VISUAL & ACTION

In the main dashboard, switch to the **"Dataset & Evidence"** tab.
Scroll through the 14 peer-reconciled US7432034B2 electron-beam records (50 keV, 6-inch silicon wafers).
Highlight evidence snippets and standardized formulation components (P-1 resin, MM-1, CL-1, PAG).

SPOKEN VOICEOVER (VO)

"High-fidelity AI requires uncompromised ground truth.

Our data foundation began with two independent researchers—Akhila and Kathe—who hand-annotated 10 semiconductor negative-resist patents.

We then deployed automated comparison techniques to audit discrepancies between the passes—resolving text-versus-table conflicts and standardized stoichiometry into a 100% consensus gold dataset of 239 records.

From this vetted foundation, we built and benchmarked two distinct implementation pathways: a hand-coded **Manual Track** and an agentic **Automated Track** developed by Antigravity."

ON-SCREEN DISPLAY / NOTES

Callout Box:
2 Human Annotators (Akhila & Kathe)
→ Automated Discrepancy Auditing
→ Reconciled 100% Consensus Gold Set

`10 Patents · 239 Records`

DEMO VIDEO STORYBOARD (PART II: AUTOMATED ANTIGRAVITY AI & COMPLIANCE)

Workflow Dimension	Manual Track (Human Evaluation & Coding)	Automated Track (Antigravity Agentic Pipeline)
Surrogate Modeling	Hand-coded parametric scripts, static response bounds	Dual GP with Matérn 5/2 kernel & LOOCV cross-validation
Bayesian Optimization	Step-by-step human inspection & manual candidate selection	Autonomous Constrained Expected Improvement with dose penalty
PFAS Screener	Checklist & manual SDS chemical lookup	Real-time programmatic OECD & REACH Annex XVII regex audit
Deployment Role	Exploratory sanity-checking & engineering boundary testing	High-throughput autonomous formulation discovery (3–4 steps)

SCENE 3: Automated Track by Antigravity — Gaussian Process & Bayesian Optimization

`1:10 - 2:00 (50 sec)`

VISUAL & ACTION

Click sidebar button: **"🎯 Run 1-Step Bayesian Optimization"**.
Watch live recommendation update to **NL-CAND-01** with predicted LWR **4.50 ± 0.15 nm**.
Switch to **"GP Surrogate Model"** tab.
Point cursor to interactive **Altair Response Surface** with blue ±1.96σ uncertainty envelope.
Highlight LOOCV metric card: **MAE 1.37 nm**.

SPOKEN VOICEOVER (VO)

"In the Automated Track, Antigravity constructed a closed-loop Gaussian Process surrogate utilizing a Matérn 5/2 covariance kernel and full epistemic uncertainty quantification.

In rigorous Leave-One-Out Cross-Validation, the model achieves a Mean Absolute Error of just 1.37 nanometers on line roughness and 0.27 microcoulombs on dose.

By maximizing Expected Improvement under stepper dose constraints, the automated engine isolates the global optimum—a dual-crosslinker synergy of 65% melamine and 35% glycoluril—in just 3 to 4 iterations, cutting experimental cycles by nearly three-fold."

ON-SCREEN DISPLAY / NOTES

Highlight Graphic:
Gaussian Process (Matérn 5/2)
Constrained Expected Improvement (EI)

`LOOCV MAE: 1.37 nm`

`3x Sample Efficiency`

SCENE 4: Automated PFAS Compliance Screening & Green Substitution

`2:00 - 2:35 (35 sec)`

VISUAL & ACTION

Switch to **"🛡️ PFAS & Regulatory Screener"** tab.
Show Legacy PAG flagged in red: *Triphenylsulfonium nonaflate (C4F9SO3-)* → *OECD PFAS Banned.*
Show Replacement PAG in green: *Triphenylsulfonium Camphorsulfonate* → *100% REACH / TSCA Clear.*
Click **"Download Candidate Recipe (JSON)"**.

SPOKEN VOICEOVER (VO)

"Optimization is useless if a formula cannot be manufactured legally.

Our automated compliance agent screens candidate chemistry against OECD PFAS inventories and EU REACH SVHC restrictions.

When legacy perfluorononaflate is detected, the agent triggers automated stoichiometric substitution with non-fluorinated triphenylsulfonium camphorsulfonate.

The bulky alicyclic camphor cage limits acid diffusion during post-exposure bake, preserving 4.5-nanometer resolution with zero fluorinated effluent."

ON-SCREEN DISPLAY / NOTES

Split Callout:
Legacy: C4F9SO3- [BANNED]
Substitute: Camphorsulfonate [PASS]

`Zero Fluorinated Effluent`

DEMO VIDEO STORYBOARD (PART III: MANUAL TRACK & OUTRO)

SCENE 5: Manual Track — Hand-Coded Evaluation & Spin Hydrodynamics

`2:35 – 3:15 (40 sec)`

VISUAL & ACTION

In sidebar, toggle workflow to: **"✍️ Manual Workbench (Chemist Calculator & Rules)"**.
Drag **Spin Speed slider** from 1000 → 1500 → 2000 RPM.
Watch live **Film Thickness Metric** recalculate via Meyerhofer model: *300 nm @ 1500 RPM*.
Demonstrate real-time validator checkmark: ✓ *Passes all mass-balance and stoichiometry rules*.

SPOKEN VOICEOVER (VO)

"To empower domain experts without relying on automated black-box loops, we developed the parallel **Manual Track**.

Here, the modeling, evaluation logic, and process calculators were hand-coded by human engineers.

Researchers can manually interrogate stoichiometric ratios, inspect total solids weight percent, and calculate spin-coating hydrodynamics in real time using Meyerhofer's evaporative equation—predicting exact film thickness at 1500 RPM.

Rule-based sanity checkers guard against improper quencher ratios or phase separation, providing complete human-in-the-loop auditability."

ON-SCREEN DISPLAY / NOTES

Track Switcher View:
Manual Modeling & Evaluation Track
Meyerhofer Film Model: $h_0 \approx 300\text{ nm}$

`Hand-Coded Validation Logic`

`Human-in-the-Loop`

SCENE 6: Technical Report, Git Repository & Conclusion

`3:15 – 3:35 (20 sec)`

VISUAL & ACTION

Briefly display the publication PDF: *Negative_Lines_Technical_Report.pdf*.
Transition to GitHub repo: *github.com/whatnotalot/negative-lines* showing commit tree.
End on title card with live Streamlit app link and open-source license.

SPOKEN VOICEOVER (VO)

"By uniting hand-annotated patent ground truth, manual engineering rigor, and Antigravity's autonomous AI optimization, Project Negative Lines establishes a validated blueprint for sustainable semiconductor manufacturing.

Explore our code, live Streamlit application, and technical paper on GitHub. Thank you!"

ON-SCREEN DISPLAY / NOTES

Final Title Screen:
GitHub: whatnotalot/negative-lines
Streamlit Community Cloud Live App

`MIT License · Fully Open Source`

PRESENTER & VIDEO CAPTURE CHECKLIST

Technical Element	Recommended Target	Presenter Note & Execution Cue
Screen Resolution	1920 × 1080 (1080p, 16:9)	Set display scaling to 100% or standard Retina; zoom browser to 110%–125% for high readability.
Sidebar Appearance	Hardcoded `#0B132B` Dark Theme	Verify text contrast and radio button icons are crisp before starting capture.
Mouse Pacing	Smooth, intentional cursor paths	Avoid erratic mouse waving; pause on KPI cards and Altair chart tooltips for 1.5 seconds.
Voiceover Pacing	~130–135 words per minute	Leave 1-second silence before and after switching tabs to facilitate clean video editing cuts.
Streamlit Command	`streamlit run app.py`	Execute from project root: `/Users/technano/agy2-projects/2026.10.OSI Hackathon/Negative Lines`.